# PRAVAAH-360: Executive Project Synopsis

Integrated AICTE Approval Workflow Automation, Compliance Monitoring, Expiry Alerts & Escalation Management System

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| **Problem Statement ID:** | SIH26262 | **Category:** | Software |
| **Sponsoring Organization:** | AICTE (Ministry of Education) | **Theme:** | Smart Automation |
| **Problem Statement Title:** | Integrated AICTE Approval Workflow Automation, Compliance Monitoring and Alert Management System | **Target TRL:** | TRL 4 / TRL 5 (Fully Integrated Prototype) |

## 1. Problem Statement Requirements & Operational Alignment

Every year, 10,500+ technical institutions apply for Extension of Approval (EoA), new program introductions, or intake expansions. The process involves high-volume multi-bureau workflows governed by the statutory Approval Process Handbook (APH) and strict Supreme Court timeline mandates (*Parshvanath Charitable Trust Case*).

**Core Operational Requirements Addressed (Directly from PS SIH26262):**
• **Statutory Document & Lease Expiry Tracking:** Real-time monitoring of institutional land leases (30-year lease expiration tracking), Fire NOC renewals, Building Stability Certificates, and FDR bank guarantee maturities with **automated 90, 60, and 30-day proactive alert notifications**.
• **Mandatory Undertakings & Affidavits Verification:** Tracking mandatory institutional affidavits (Anti-Ragging undertakings, Faculty Cadre Ratio compliance, Fee Regulatory compliance) with automated deficiency flags.
• **Stage-Wise Sequential Routing:** Clear stage progression: Online Application $\rightarrow$ Automated Document & APH Scrutiny $\rightarrow$ EVC Field Inspection $\rightarrow$ Regional Committee Review $\rightarrow$ Standing Appellate Committee (SAC) $\rightarrow$ Final Officer Sign-Off.
• **Bureau-Wise Pendency Dashboard:** Real-time analytics tracking file pendency, stage bottlenecks, and active application states across AICTE Bureaus and 8 Regional Offices.
• **Hierarchy-Based Automated Escalation Engine:** Configurable SLA triggers (e.g. Scrutiny SLA: 7 days). If an assigned officer or committee delays action, the file automatically escalates: Assigned Officer $\rightarrow$ Regional Officer (RO) $\rightarrow$ Bureau Head $\rightarrow$ Member Secretary.
• **Human-in-the-Loop Officer Decision Support:** The system assists with automated deficiency calculations and audit trails, while the **final approval or rejection decision rests strictly with authorized AICTE officers**.
• **Seamless Integration with Existing AICTE Systems:** RESTful API bridges connecting with the existing AICTE Web Portal, DigiLocker, Parakh/NAD, and National SMS/Email Gateways.

## 2. The PRAVAAH-360 Solution Architecture

PRAVAAH-360 is engineered as an enterprise-grade, realistic, and fully compliant workflow automation system built on four functional pillars:

### Pillar 1: Stage-Wise Approval Workflow & Rules Engine

A robust state-machine workflow engine that validates institutional submissions against APH formulas (land carpet area, student-faculty ratio, lab requirements) in seconds. It verifies uploaded institutional undertakings, land lease tenure, and statutory certificates, auto-generating instant deficiency scorecards and cure notices.

### Pillar 2: Native Android EVC Mobile Inspection Companion

A native Android application (Kotlin/Flutter) for visiting committee experts with hardware root-of-trust: utilizes Android's Location.isFromMockProvider() API to strictly detect and block GPS spoofing, captures live camera photos of campus infrastructure with embedded GPS EXIF metadata, and supports offline queue caching with automatic sync.

### Pillar 3: Expiry Alert Cascade & Multi-Tier Escalation Engine

A background daemon engine (Celery + Redis) continuously evaluates statutory document expiry dates (land lease, Fire NOC, FDR) and operational SLAs. It sends proactive multi-channel alerts (SMS/Email) to institutions and automatically escalates overdue

files through AICTE administrative tiers.

### Pillar 4: Bureau Pendency Dashboard & Append-Only Audit Logging

A unified command console for AICTE leadership and Regional Directors providing live heatmap visibility into pendency queues. All workflow actions, officer remarks, and document changes are recorded in an append-only PostgreSQL audit log backed by Digital Signature Certificates (DSC) for court non-repudiation.

## 3. System Flowcharts & Procedural Pipelines

The following diagrams illustrate the concrete procedural pipelines of PRAVAAH-360, demonstrating stage-wise approval routing, automated expiry alerts, native mobile inspection, pendency heatmaps, and hierarchical escalations.

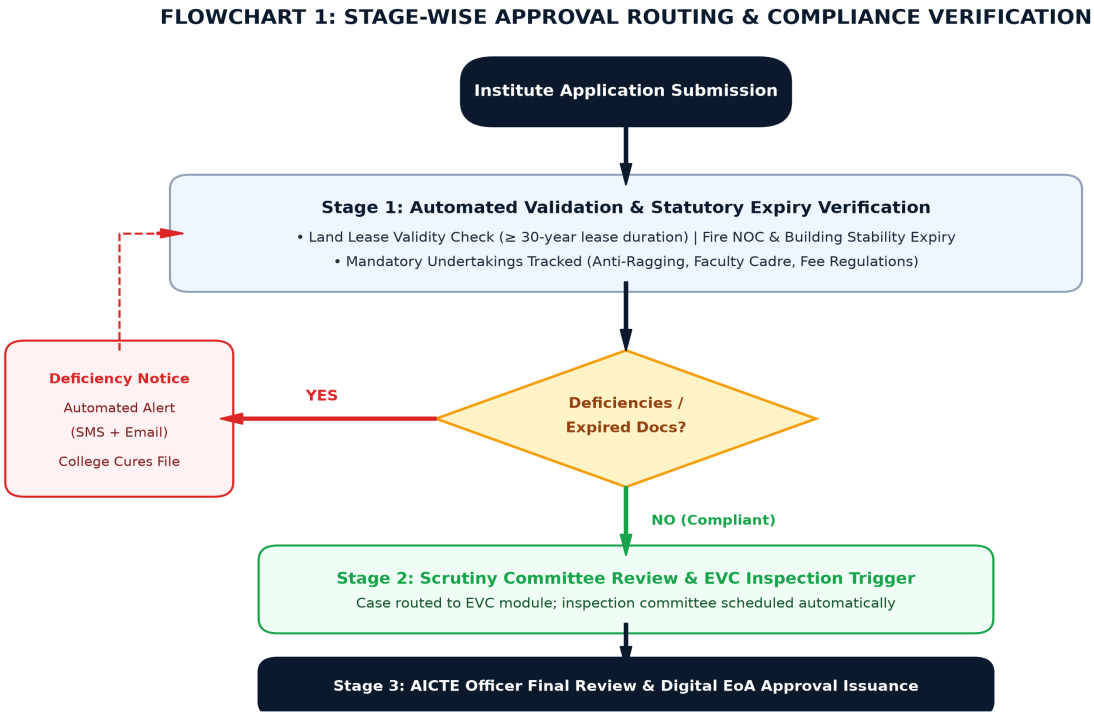



*Figure 1: Stage-Wise Application Routing, Document & Lease Expiry Verification, and Deficiency Remediation Pipeline.*

## 3. System Flowcharts (Continued: EVC Inspection & Escalation)

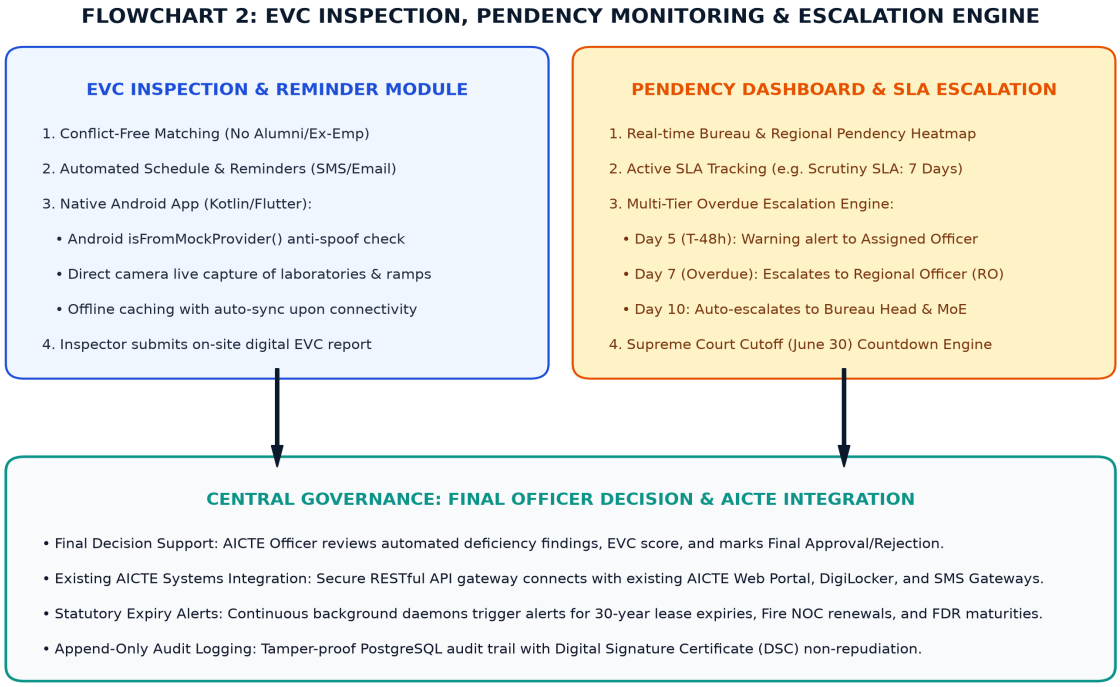



*Figure 2: Native Android EVC App Verification, Bureau Pendency Dashboard, and Multi-Tier Overdue Escalation Engine.*

## 4. Technical Architecture & Technology Stack

| Layer | Components & Technologies | Operational Function |
|---|---|---|
| **Presentation & Admin** | Next.js 14, React 18, Tailwind CSS, Shadcn UI | Institution web portal, AICTE Bureau pendency dashboard, officer scrutiny console. |
| **Field Mobility (EVC)** | Native Android (Kotlin / Flutter), Room SQLite | Anti-mock GPS verification (isFromMockProvider), live camera capture, offline sync. |
| **API Gateway & Auth** | FastAPI, Nginx, OAuth 2.0, DigiLocker Gateway | Secure RESTful APIs connecting to existing AICTE databases and DigiLocker repositories. |
| **Rules & Workflows** | Python Rule Engine (PyDrools/JSON-e), Celery Workers | Statutory APH rule validation, stage-wise state machine, automated SLA escalations. |
| **Persistence & Security** | PostgreSQL 16 (Append-Only Audit), MinIO S3, DSC | **ACID database, encrypted document vault, Digital Signature Certificate (DSC) support.** |

## 5. Feasibility, TRL Progression & Risk-Mitigation Matrix

**TRL Progression:** The solution is currently at **TRL 3** and will be demonstrated at **TRL 4 / TRL 5** during the hackathon finale with end-to-end integration across mock AICTE institution records, automated expiry daemons, and working native Android APK builds.

| Identified Challenge | Severity | Operational Risk / Impact | Technical Mitigation Strategy |
|---|---|---|---|
| **Mock GPS & Fake Campus Inspections** | **HIGH** | Inspectors uploading pre-saved fake photos from off-campus locations. | **Native Android Location API:** Uses isFromMockProvider() to reject fake location apps; enforces live camera capture. |

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| **Inspector Collusion & Pre-Visit Lobbying** | **HIGH** | Institutes discovering visiting committee identity and attempting lobbying. | **Blind Automated Dispatch:** Inspector assignments are hidden until T-24h; automated SMS alerts dispatched concurrently. |
| **Land Lease & Statutory Expiry Lapses** | MEDIUM | Institutions operating on lapsed land leases or expired fire NOCs. | **Proactive Expiry Engine:** Automated 90/60/30-day alert cascade triggers compliance notices before approval window. |
| **Bureau Pendency & Deadlines Breach** | **CRITICAL** | Bottlenecks causing failure of Supreme Court June 30 approval cutoffs. | **Multi-Tier Escalation Engine: Overdue files automatically escalate to Regional Director and Bureau Head.** |
| **Integration with Existing AICTE Systems** | MEDIUM | Data sync failures between new modules and existing AICTE database. | **Standardized RESTful APIs:** Decoupled microservices architecture with OpenAPI specifications for zero disruption. |

## 6. Quantifiable Impact & Beneficiary Value

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| **75% Turnaround Reduction** Reduces approval processing from 90 days to < 20 days. | **100% Expiry Compliance** Zero unmonitored land lease or fire safety certificate lapses. | **Zero Overdue Bottlenecks** Hierarchy-based auto-escalation clears file pendency. | **100% Officer Accountability** Tamper-proof DSC audit trail for court defense. |

**Multistakeholder Value Matrix:**

• **AICTE Officers & Bureau Leadership:** Clear pendency visualization across regions, automated reminders, and complete decision authority.
• **10,500+ Technical Institutions:** Timely alerts for document & lease renewals, transparent stage tracking, and instant deficiency feedback.
• **Visiting Committee Experts (EVC):** Intuitive native mobile inspection app with offline caching and automated report compilation.
• **Students & Citizens:** Assured safety and accreditation compliance (valid fire NOC, 30-year land lease, verified faculty cadre).

## 7. Statutory References & Deliverable Proofs

• **Supreme Court of India:** *Parshvanath Charitable Trust & Ors. v. AICTE (2013) 3 SCC 385* (Annual Approval Calendar Mandate).
• **AICTE Statutory Regulations:** Approval Process Handbook (APH 2024–2027) Norms for Land, Cadre Ratios, and Affidavits.
• **MeitY Guidelines:** Guidelines for Compliance of Digital Signature Certificates (DSC) and e-Sign in Government Applications.
• **Live Prototype & Deliverables:**
- Production Web Application: https://frontend-lyart-chi-21.vercel.app
- GitHub Repository: https://github.com/divyanshattrish/pravaah360-aicte.git
- Demonstration Folder: https://drive.google.com/drive/folders/1Pravaah360DemoSIH2026?usp=sharing